INCH-POUND

MIL-C-85049/31B
30 November 1990
SUPERSEDING
MIL-C-0085049/31A(USAF)
21 November 1986
MIL-C-85049/31
16 January 1980
MIL-C-83723/15
7 May 1976
MS3416F
10 May 1982

MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, BACKSHELL, NONENVIRONMENTAL, STRAIGHT, NON-SELF-LOCKING AND SELF-LOCKING, CATEGORY 3B (FOR MIL-C-5015 CRIMP, MIL-C-26482 SERIES 2, MIL-C-81703 SERIES 3, AND MIL-C-83723 SERIES III CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-85049.

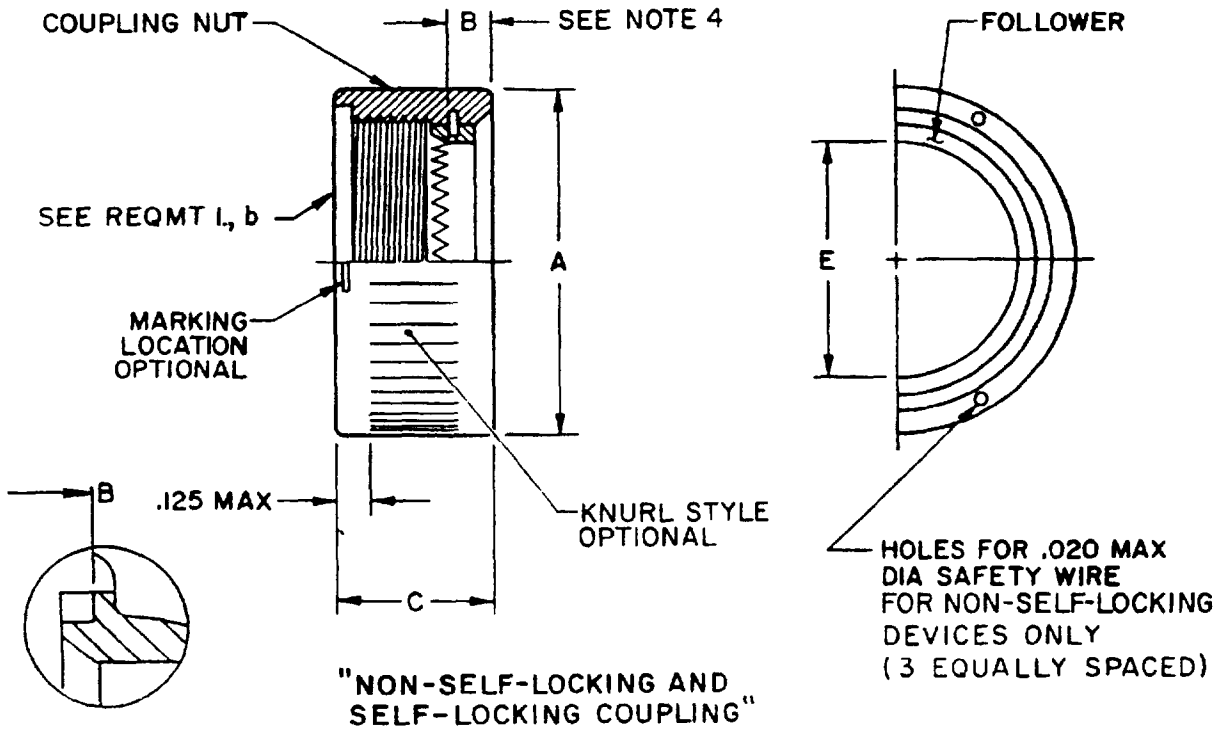


FIGURE 1. Configuration and dimensions.

AMSC N/A
FSC 5935
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TABLE I. Dash numbers, shell size and dimensions.

Dash Number	For Connection Shell Size (Ref)				Non-Self-Locking				Self-Locking			
	MIL-C-81703 Series 3	MIL-C-26482 Series 2	MIL-C-5015 Crimp	MIL-C-83723 Series III	A Max Dia	B Max	C Max	E Max	A Max Dia	B Max	C Max	E Max
-3	3				.669	.220	.540	.270				
-8		8	8S	8	.617	↓	↓	.270	.885	.39	.710	.270
-10		10	10S,10SL	10	.734			.375	1.010	↓	↓	.375
-12	7	12	12S&12	12	.858			.511	1.135			.511
-14	12	14	14S&14	14	.984			.585	1.260			.585
-16	19	16	16S&16	16	1.112			.710	1.385			.710
-18	27	18	18	18	1.218			.789	1.510			.784
-20	37	20	20	20	1.345			.914	1.635			.914
-22		22	22	22	1.468		↓	1.039	1.760	↓	↓	1.039
-24		24	24	24	1.593		.540	1.154	1.885	.39	710	1.154
-28			28		1.969		.702	1.389	2.135	.41	.890	1.389
-32			32		2.219		↓	1.635	2.395	↓	↓	1.635
-36			36		2.469			1.850	2.635			1.850
-40			40		2.719			2.065	2.885			2.065
-44			44		2.969		↓	2.320	3.135	↓	↓	2.320
-46			48		3.219	↓	.702	2.570	3.385	.41	890	2.570
-61	61				1.653	.220	.540	1.194				

INCH	mm	INCH	mm	INCH	mm
0.010	0.25	0.734	18.64	1.635	41.53
0.020	0.51	0.784	19.91	1.653	41.53
0.030	0.76	0.789	20.04	1.760	44.70
0.045	1.14	0.858	21.79	1.850	46.99
0.062	1.57	0.885	22.48	1.885	47.88
0.125	3.18	0.890	22.60	1.969	50.01
0.220	5.58	0.914	23.22	2.065	52.45
0.250	6.35	0.984	24.99	2.135	54.23
0.270	6.86	1.010	25.65	2.219	56.36
0.320	8.13	1.039	26.39	2.320	58.93
0.340	8.64	1.112	28.24	2.395	60.83
0.375	9.53	1.135	28.83	2.469	62.71
0.380	9.65	1.154	29.31	2.570	65.28
0.390	9.90	1.194	30.33	2.615	66.42
0.410	10.41	1.218	30.94	2.635	66.93
0.511	12.98	1.260	32.00	2.719	69.06
0.540	13.72	1.345	34.16	2.885	73.28
0.585	14.86	1.385	35.18	2.969	75.41
0.617	15.67	1.389	35.28	3.135	79.63
0.669	16.99	1.468	37.29	3.219	81.76
0.702	17.83	1.510	38.35	3.385	85.98
0.710	18.03	1.593	40.46		

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REQUIREMENTS:

1. Design and construction:

   a. Dimensions and configuration: See figure 1 and table I.

   b. Interface dimensions shall conform to MIL-C-85049 figure 4.

   c. The accessory shall consist of a coupling nut and follower. The coupling nut shall be captivated to and free to rotate on the follower.

2. Material and finish: See table II.

3. Accessories will not accommodate connectors using size 8, 4 or 0 contacts.

TABLE II. Material and finish.

Figure	Material	Finish
1	Aluminum alloy in accordance with MIL-C-85049	1/ N, 2/ W A

1/ Not for Navy use. Restricted to Air Force Space applications only.
2/ W finish is not for use in space application.

4. Qualification:

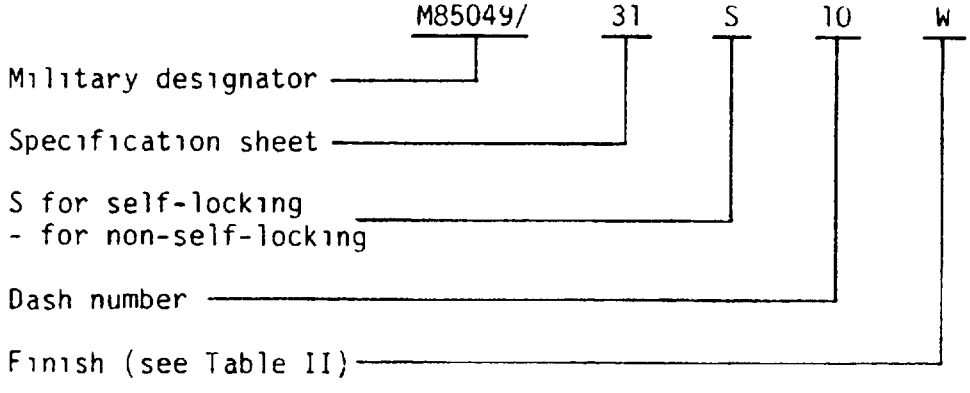
   Non-self-locking and self-locking -

   See MIL-C-85049, category 3B. Bending moment, cable pull-out and shell conductivity tests not applicable.

   Information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Avionics Center, Attention: Code B/444, 6000 East 21st Street, Indianapolis, IN 46219-2189.

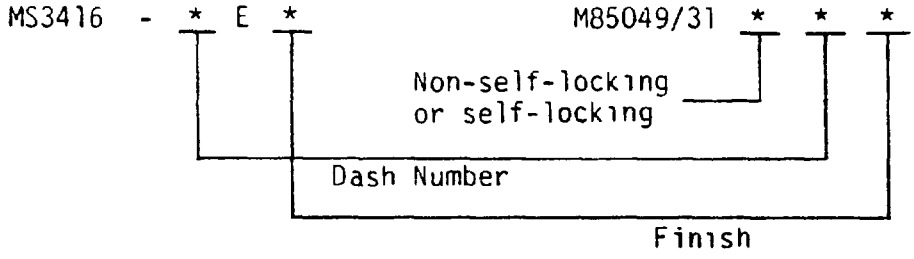
5. Self-locking devices shall be in accordance with MIL-C-85049.

6. Part Number: See example below:



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7. Supersession data:



Old	New
A	W
C	W
G	A
N	N

NOTES:

1. Dimensions are in inches.

2. Metric equivalents are based upon 1.00 inch = 25.4 mm and shall be in accordance with ASTM E380, paragraph 4.5, Method A and ASTM E29, rounding-off method.

3. Dimensions apply after plating.

4. Measured from the root of the tooth to the back of the coupling nut.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11, 15, 19, 99
DLA - ES

Preparing activity.
Navy - AS
Project No. 5935-3734-01

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